

Deyun Lyu

Research Fellow | National Institute of Informatics

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About Me

I am a Research Fellow at the National Institute of Informatics, working in the Trustworthy & Smart Software Engineering Lab led by Prof. **Fuyuki Ishikawa**. I received my Ph.D. in Information Science and Electrical Engineering from Kyushu University under the supervision of Prof. **Jianjun Zhao**, with co-advisors Prof. **Paolo Arcaini** and Prof. **Zhenya Zhang**. My research interests focus on the quality assurance of AI-enabled cyber-physical systems (AI-CPSS).

Research Interests

- Trustworthy AI-Enabled Cyber-Physical Systems
- Safe and Reliable Autonomous Driving Systems
- Software Engineering for AI Systems (SE4AI)
- Formal Verification and Validation

Education

Kyushu University

Ph.D. in Advanced Information Technology

Supervisor: Prof. Jianjun Zhao, Co-advisors: Prof. Paolo Arcaini, Prof. Zhenya Zhang

Fukuoka, Japan

Oct. 2020 – Mar. 2025

Graduated

Dalian University of Technology

M.S. in Software Engineering

Supervisor: Prof. Weiqiang Kong

Dalian, China

Sept. 2018 – June 2020

Graduated

Dalian University of Technology

B.S. in Network Engineering

Supervisor: Prof. Jing Gao

Dalian, China

Sept. 2014 – June 2018

Graduated

Work Experience

National Institute of Informatics

Information Systems Architecture Science Research Division

Research Fellow

Tokyo, Japan

Apr. 2025 – Present

Kyushu University

Graduate School of Information Science and Electrical Engineering

Teaching Assistant

Course: *Python Programming Exercises* for undergraduate students

Fukuoka, Japan

Summer 2024

National Institute of Informatics

Information Systems Architecture Science Research Division

Tokyo, Japan

<i>Visiting Researcher</i>	<i>Sept. 2023 – Mar. 2024</i>
Kyushu University	Fukuoka, Japan
<i>Graduate School of Information Science and Electrical Engineering</i>	
<i>Teaching Assistant</i>	<i>Summer 2023</i>
Course: <i>Python Programming Exercises</i> for undergraduate students	
Japan Science and Technology Agency	Fukuoka, Japan
<i>Research Fellow (JST-SPRING)</i>	<i>Oct. 2021 – Sept. 2023</i>
Kyushu University	Fukuoka, Japan
<i>Graduate School of Information Science and Electrical Engineering</i>	
<i>Research Assistant</i>	<i>Oct. 2020 – Sept. 2021</i>
Dalian University of Technology	Dalian, China
<i>School of Software</i>	
<i>Teaching Assistant</i>	<i>Fall 2019</i>
Course: <i>C++ Programming Exercises</i> for undergraduate students	
Dalian University of Technology	Dalian, China
<i>School of Software</i>	
<i>Teaching Assistant</i>	<i>Spring 2019</i>
Course: <i>C Programming Exercises</i> for undergraduate students	
Dalian University of Technology	Dalian, China
<i>School of Software</i>	
<i>Teaching Assistant</i>	<i>Fall 2018</i>
Course: <i>Operating Systems Practicum</i> for undergraduate students	
Dalian University of Technology	Dalian, China
<i>School of Software</i>	
<i>Research Assistant</i>	<i>Sept. 2018 – June. 2020</i>

Publications

- Quality Assurance of AI-Enabled Cyber-Physical Systems**
- **[AI-CPS-P8] Deyun Lyu**, Paolo Arcaini, Zhenya Zhang, Masaaki Inoue, Nayuta Yanagisawa, Ryo Neyama, Shintaro Fukushima and Fuyuki Ishikawa. “On the Effectiveness of Open-Loop DNN Repair in End-to-End Autonomous Driving Systems.” IEEE International Symposium on Software Reliability Engineering. (ISSRE 2026, under review, [an industry – academia collaboration project with Toyota Motor Corporation](#))
 - **[AI-CPS-P7] Deyun Lyu**, Jiayang Song, Zhenya Zhang, Zhijie Wang, Tianyi Zhang, Lei Ma and Jianjun Zhao. “Repair of DNN Control Policies via STL-Guided Patch Synthesis.” IEEE Transactions on Reliability. (TREL 2025, under review)
 - **[AI-CPS-P6] Deyun Lyu**, Zhenya Zhang, Paolo Arcaini, Thomas Laurent, Borui Zhang, Fuyuki Ishikawa, and Jianjun Zhao. “ContrRep: Search-Based Repair of DNN Controllers of AI-Enabled CPSs.” Automated Software Engineering. (ASEJ 2025, under review)
 - **[AI-CPS-P5] Deyun Lyu**, Yi Li, Zhenya Zhang, Paolo Arcaini, Xiao-Yi Zhang, Fuyuki Ishikawa and Jianjun Zhao. “Fault Localisation of AI-Enabled Cyber-Physical Systems by Exploiting Temporal Neuron Activation.” Journal of Systems and Software. (JSS 2025)
 - **[AI-CPS-P4] Deyun Lyu**, Zhenya Zhang, Paolo Arcaini, Xiao-Yi Zhang, Fuyuki Ishikawa and Jianjun Zhao. “SpectAcle: Fault Localisation of AI-Enabled CPS by Exploiting Sequences of DNN Controller Inferences.” ACM Transactions on Software Engineering and Methodology. (TOSEM

2025, **IPJS SIG-SE Excellent Research Award**)

- **[AI-CPS-P3]** Deyun Lyu, Zhenya Zhang, Paolo Arcaini, Fuyuki Ishikawa, Thomas Laurent and Jianjun Zhao. "Search-Based Repair of DNN Controllers of AI-Enabled CPS Guided by System-Level Specifications." Proceedings of the Genetic and Evolutionary Computation Conference. (GECCO 2024)
- **[AI-CPS-P2]** Zhenya Zhang, Deyun Lyu, Paolo Arcaini, Lei Ma, Ichiro Hasuo and Jianjun Zhao. "FalsifAI: Falsification of AI-Enabled Hybrid Control Systems Guided by Time-Aware Coverage Criteria." IEEE Transactions on Software Engineering. (TSE 2023)
- **[AI-CPS-P1]** Jiayang Song, Deyun Lyu*, Zhenya Zhang, Zhijie Wang, Tianyi Zhang and Lei Ma. "When Cyber-Physical Systems Meet AI: A Benchmark, an Evaluation, and a Way Forward." 44th International Conference on Software Engineering: Software Engineering in Practice. (ICSE-SEIP 2022, **Co-first author**)

Quality Assurance of Cyber-Physical Systems.....

- **[CPS-P4]** Tanmay Khandait, Deyun Lyu*, (12 other authors). "ARCH-COMP 2025 Category Report: Falsification." 12th International Workshop on Applied Verification of Continuous and Hybrid Systems. (**Co-first author**)
- **[CPS-P3]** Yipei Yan, Deyun Lyu, Zhenya Zhang, Paolo Arcaini and Jianjun Zhao. "Automated Generation of Benchmarks for Falsification of STL Specifications." IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems. (TCAD 2025)
- **[CPS-P2]** Zhenya Zhang, Deyun Lyu, Paolo Arcaini, Lei Ma, Ichiro Hasuo and Jianjun Zhao. "Effective Hybrid System Falsification Using Monte Carlo Tree Search Guided by QB-Robustness." 33rd International Conference on Computer-Aided Verification. (CAV 2021)
- **[CPS-P1]** Zhenya Zhang, Deyun Lyu, Paolo Arcaini, Lei Ma, Ichiro Hasuo and Jianjun Zhao. "On the Effectiveness of Signal Rescaling in Hybrid System Falsification." 13th NASA Formal Methods Symposium. (NFM 2021)

Quality Assurance of Autonomous Driving Systems.....

Note: These works were conducted during my time at Dalian University of Technology. While my official academic name is "Deyun Lyu", one of them ([ADS-Verf-P1]) lists me as "Deyun Lv" following pinyin conventions.

- **[ADS-Verf-P2]** Huihui Wu, Deyun Lyu, Y. Zhang, Gang Hou, Masahiko Watanabe, Jie Wang and Weiqiang Kong. "A Verification Framework for Behavioural Safety of Self-Driving Cars." IET Intelligent Transport Systems.
- **[ADS-Verf-P1]** Huihui Wu, Deyun Lyu, TengXiang Cui, Gang Hou, Masahiko Watanabe and Weiqiang Kong. "SDLV: Verification of Steering Angle Safety for Self-Driving Cars." Formal Aspects of Computing. (FAC 2021)

Paolo, Michael, Fuyuki and then Cristina, Alessandro, Deyun.

Human – Robot Interaction.....

- **[HRI-P1]** Massimo Donini, Paolo Arcaini, Michael Oliverio, Fuyuki Ishikawa, Alessandro Mazzei, Deyun Lyu, Cristina Gena. "Coordinating Speech with Touch Input and Visual Cues in Human – Robot Interaction: A Multimodal System Evaluated through Metamorphic Testing." 21st Annual IEEE/ACM International Conference on Human-Robot Interaction. (HRI 2026)

Invited Talks, Presentations, and Posters

Invited Talks.....

- **[IT3]** “Quality Assurance of AI-Enabled Cyber-Physical Systems.” The 220th IPSJ/SIGSE Software Engineering Workshop, Todaiji Culture Center, Nara, Japan. (Nov. 2025)
- **[IT2]** “SpectAcle: Fault Localisation of AI-Enabled CPS by Exploiting Sequences of DNN Controller Inferences.” IPSJ/SIGSE Software Engineering Symposium, Waseda University, Tokyo, Japan. (Sept. 2025)
- **[IT1]** “Search-Based Repair of DNN Controllers of AI-Enabled Cyber-Physical Systems Guided by System-Level Specifications.” Top Conference/Journal Special Talk, 41st Annual Conference of the Japan Society for Software Science and Technology, Ritsumeikan University, Osaka, Japan. (Sept. 2024)

Presentations.....

- **[Pres4]** “SpectAcle: Fault Localisation of AI-Enabled CPS by Exploiting Sequences of DNN Controller Inferences.” Journal-First Track, 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025), Seoul, South Korea. (Nov. 2025)
- **[Pres3]** “SSBSE Summary of Search-Based Repair of DNN Controllers of AI-Enabled Cyber-Physical Systems Guided by System-Level Specifications.” Hot-off-the-Press Track, 17th Symposium on Search-Based Software Engineering (SSBSE 2025), Seoul, South Korea. (Nov. 2025)
- **[Pres2]** “Search-Based Repair of DNN Controllers of AI-Enabled CPS Guided by System-Level Specifications.” Proceedings of the Genetic and Evolutionary Computation Conference (GECCO 2024), Melbourne, Australia. (July 2025)
- **[Pres1]** “Towards Building Reliable AI-Enabled Cyber-Physical Systems.” Doctoral Symposium, 26th International Symposium on Formal Methods (FM 2023), Lübeck, Germany. (Mar. 2023, Accepted, but not presented due to COVID-19 travel restrictions.)

Posters.....

- **[Poster1]** “FalsifAI: Falsification of AI-Enabled Hybrid Control Systems Guided by Time-Aware Coverage Criteria.” Demo/Poster Session, 39th Annual Conference of the Japan Society for Software Science and Technology, Nanzan University, Nagoya, Japan. (Sept. 2022)

Patents

Note: These patents were filed during my time at Dalian University of Technology. In U.S. patents, my name appears as “Deyun Lv”, following pinyin conventions. In Chinese patents, my name appears as “吕德云”. My official academic name is “Deyun Lyu”.

U.S. Patents.....

- **[US-Pt3]** Weiqiang Kong, **Deyun Lyu**, Zhong Wei, Risheng Liu, Xin Fan and Zhongxuan Luo. “Method For Infrared Small Target Detection Based On Depth Map In Complex Scene.” US Patent, No. 12,108,022, Oct. 2024. (Granted, Serve as the primary contributor)
- **[US-Pt2]** Weiqiang Kong, **Deyun Lyu**, Zhong Wei, Risheng Liu, Xin Fan and Zhongxuan Luo. “Method for Fully Automatically Detecting Chessboard Corner Points.” US Patent, No. 12,094,152, Sept. 2024. (Granted, Serve as the primary contributor)
- **[US-Pt1]** Zhong Wei, **Deyun Lyu**, Weiqiang Kong, Risheng Liu, Xin Fan, Zhongxuan Luo and Shengquan Li. “Method for Adaptively Detecting Chessboard Sub-Pixel Level Corner Points.” US Patent, No. 11,900,634, Feb. 2024. (Granted, Serve as the primary contributor)

Chinese Patents.....

- **[CN-Pt6]** Weiqiang Kong, **Deyun Lyu**, Wei Zhong, Risheng Liu, Xin Fan, Zhongxuan Luo. “A Fully Automatic Checkerboard Corner Detection Method”, CN Patent, No. CN111243032B, May 2023. (Granted, Serve as the primary contributor)
- **[CN-Pt5]** Weiqiang Kong, **Deyun Lyu**, Wei Zhong, Risheng Liu, Xin Fan, Zhongxuan Luo. “An Infrared Small Target Detection Method Based on Depth Map in Complex Scenes”, CN Patent, No. CN111209877B, May 2023. (Granted, Serve as the primary contributor)
- **[CN-Pt4]** Wei Zhong, **Deyun Lyu**, Weiqiang Kong, Risheng Liu, Xin Fan, Zhongxuan Luo, Shengquan Li. “A Checkerboard Sub-pixel Level Corner Adaptive Detection Method”, CN Patent, No. CN111260731B, May 2023. (Granted, Serve as the primary contributor)
- **[CN-Pt3]** Wei Zhong, **Deyun Lyu**, Weiqiang Kong, Risheng Liu, Xin Fan, Zhongxuan Luo, Shengquan Li. “A Color Small Target Detection Method Based on Depth Map in Complex Scenes”, CN Patent, No. CN111291747B, June 2023. (Granted, Serve as the primary contributor)
- **[CN-Pt2]** **Deyun Lyu**, Wei Zhong, Weiqiang Kong, Risheng Liu, Xin Fan, Zhongxuan Luo. “Method for Quickly Registering Visible Light and Infrared Camera”, CN Patent, No. CN111242991B, Sept 2022. (Granted)
- **[CN-Pt1]** Wei Zhong, **Deyun Lyu**, Weiqiang Kong, Risheng Liu, Xin Fan, Zhongxuan Luo, Shengquan Li. “Method for Optimizing External Parameters of Binocular Camera”, CN Patent, No. CN111243033B, Sept 2022. (Granted, Serve as the primary contributor)

Professional Memberships and Services

Academic Memberships.....

- Member, IEEE (Institute of Electrical and Electronics Engineers)
- Member, IPSJ (Information Processing Society of Japan)
- Member, IPSJ SIG-SE (Special Interest Group on Software Engineering)

Professional Services.....

Journal Paper

- Reviewer, Science of Computer Programming (SCP 2026)
- Reviewer, Automated Software Engineering (ASEJ 2026)
- Reviewer, ACM Transactions on Software Engineering and Methodology (TOSEM 2025)
- Reviewer, Empirical Software Engineering (EMSE)
- Reviewer, IEEE Transactions on Reliability (TRel)
- Reviewer, Research Directions: Cyber-Physical Systems

Conference Paper

- PC Member of Tools and Datasets Track, ASE 2026
- PC Member of ECACPS Workshop, GECCO 2026
- PC Member of Hot-off-the-Press Track, SSBSE 2026
- Web Chair and Artifact Evaluation PC Member, International Symposium on Formal Methods (FM 2026)

- PC Member of Artifact Evaluation Track, International Symposium on Automated Technology for Verification and Analysis (ATVA 2025)
- PC Member, Association for the Advancement of Artificial Intelligence (AAAI 2023, 2024, 2025)
- Reviewer, International Conference on Software Engineering (ICSE)
- Reviewer, IEEE/ACM International Conference on Automated Software Engineering (ASE)
- Reviewer, ACM SIGSOFT Symposium on the Foundations of Software Engineering (FSE)
- Reviewer, International Symposium on Software Testing and Analysis (ISSTA)
- Reviewer, International Conference on Machine Learning (ICML)
- Reviewer, International Conference on Embedded Software (EMSOFT)
- Reviewer, IEEE International Conference on Software Testing, Verification and Validation (ICST)
- Reviewer, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- Reviewer, International Conference on Engineering of Complex Computer Systems (ICECCS)
- Reviewer, International Symposium on Theoretical Aspects of Software Engineering (TASE)
- Reviewer, International Conference on AI Engineering – Software Engineering for AI (CAIN)

Other Research Experience

Quality Assurance of End-to-End Autonomous Driving Systems.....

National Institute of Informatics

Tokyo, Japan

Researcher

July 2025 – Present

Collaboration with Toyota, JARI and GAIO.

- Demonstrated the potential of my prior research on end-to-end ADS.
- Developing new fault localisation and repair techniques for large-scale end-to-end controllers.

Testing of Multimodal Social Robots.....

National Institute of Informatics

Tokyo, Japan

Researcher

June 2025 – Present

Collaboration with University of Turin.

- Ongoing development of a multimodal-aware metamorphic testing method of social robots (with Pepper robot as a case study).

eBPF-assisted Falsification of Temporal Logic Specifications under Noisy Environments.....

National Institute of Informatics

Tokyo, Japan

Researcher

Apr. 2025 – Present

Collaboration with Kyushu University and University of Bergamo.

- Ongoing development of an eBPF-assisted falsification framework under noisy environments.

Benchmark Contribution to the 2025 ARCH Falsification Competition.....

National Institute of Informatics

Tokyo, Japan

Researcher

Apr. 2025 – June. 2025

ARCH-COMP is a well-established, internationally recognized competition in CPS community.

- Developed a novel benchmark generation method for falsification[CPS-P3] and contributed five benchmarks to the 2025 ARCH Falsification Competition.
- Participated in the competition with our falsification tool *ForeSee*[CPS-P2].

Engineerable AI Project.....

National Institute of Informatics

Tokyo, Japan

Visiting Researcher

Sept. 2023 – Mar. 2024

University research project funding by JST Mirai project (Grant No. JPMJMI20B8).

- Developed a fault localisation approach to localise faulty weights in DNN controllers.
- Proposed the first search-based repair approach for DNN controllers of AI-CPSs.

Testing, Analysis, and Repair of AI-Enabled Cyber-Physical Systems.....

Kyushu University

Fukuoka, Japan

Research assistant

Apr. 2023 – Mar. 2025

University research project funded by JSPS (Grant No. 23H03372).

- Designed an approach to localise faulty neurons in DNN controllers of AI-CPSs.
- Developed a data-driven method for automatically generating synthetic benchmarks for falsification.

機械がバグを修正する時代—擬似オラクル生成・適用と自動バグ修正技術の深化.....

Kyushu University

Fukuoka, Japan

Research assistant

Jan. 2021 – Aug. 2022

University research project funded by JSPS (Grant No. 21H04877).

- Proposed eight time-aware coverage criteria and developed FalsifAI, the first coverage-guided falsification framework for AI-CPSs.

Comprehensive Analysis and Repairing Techniques for Stateful Deep Learning Systems.....

Kyushu University

Fukuoka, Japan

Research assistant

May 2021 – Jan. 2022

University research project funded by JSPS (Grant No. 20H04168).

- Established the first public benchmarks for AI-CPSs, enabling systematic evaluation and guiding testing research.

Traffic Accident Scene Reconstruction and Preprocessing Using TLS Point Clouds .

Dalian University of Technology

Dalian, China

Research Assistant

Sept. 2019 – June. 2020

Collaboration with Guangzhou Traffic Police Department.

- Developed a 3D point cloud-based system for traffic accident scene investigation and preprocessing.
- Implemented the system as a practical tool and deployed it in practice.

Vehicle-Mounted Visible – Infrared Stereo Vision Perception Unit.....

Dalian University of Technology

Dalian, China

Research Assistant

June 2018 – Sept. 2019

University – Industry Collaboration Project, resulting in multiple US and CN patents (See Section Patents).

- Designed automated sub-pixel chessboard corner detection methods and implemented them as practical calibration tools.
- Developed efficient external parameter optimization and registration approaches for visible – infrared stereo vision perception systems.
- Proposed a depth map-based method for detecting small infrared/colour targets in complex scenes.

Teaching Assistantships

- **Python Programming Exercises**, Summer 2024, Kyushu University, Fukuoka, Japan (approx. 60 undergraduate students)

- **Python Programming Exercises**, Summer 2023, Kyushu University, Fukuoka, Japan (approx. 30 undergraduate students)
- **C++ Programming Exercises**, Fall 2019, Dalian University of Technology, Dalian, China (approx. 100 undergraduate students)
- **C Programming Exercises**, Spring 2019, Dalian University of Technology, Dalian, China (approx. 100 undergraduate students)
- **Operating Systems Practicum**, Fall 2018, Dalian University of Technology, Dalian, China (approx. 100 undergraduate students)

Mentoring Experience

- **Chenchang Wang**, Expected to graduate in March 2027, Kyushu University, Fukuoka, Japan
Master's thesis: *eBPF-assisted Falsification of Temporal Logic Specifications under Noisy Environments*
- **Borui Zhang**, Expected to graduate in March 2026, Kyushu University, Fukuoka, Japan
Master's thesis: *ContrRep: Search-Based Repair of DNN Controllers of AI-Enabled CPSs*
- **Yipei Yan**, Graduated in March 2025, Kyushu University, Fukuoka, Japan
Master's thesis: *Automated Generation of Benchmarks for Falsification of STL Specifications*
- **Yi Li**, Graduated in March 2025, Kyushu University, Fukuoka, Japan
Master's thesis: *Tactical: Fault Localization of AI-Enabled Cyber-Physical Systems by Exploiting Temporal Neuron Activation*
- **Ruinan Zhang**, Graduated in June 2022, Dalian University of Technology, Dalian, China
Master's thesis: *Research on Test Prioritization Technique for Deep Neural Networks*

Research Grants and Funding

- Principal Investigator, Support for Pioneering Research Initiated by the Next Generation, Japan Science and Technology Agency (JST), Oct. 2021 – Sept. 2023. Grant No. JPMJSP2136. Amount: JPY 500,000 annually (total JPY 1,000,000).

Awards

2025 IPSJ SIG-SE Excellent Research Award Aug. 2025
Information Processing Society of Japan, Special Interest Group on Software Engineering (IPSJ SIGSE)

For our TOSEM 2025 paper entitled "SpectAcle: Fault Localisation of AI-Enabled CPS by Exploiting Sequences of DNN Controller Inferences."

2025 IPSJ SIG-SE Recommended Doctoral Dissertation June 2025
Information Processing Society of Japan, Special Interest Group on Software Engineering (IPSJ SIGSE)

GECCO 2024 Student Travel Grant May 2024
ACM Special Interest Group on Genetic and Evolutionary Computation (ACM SIGEVO)
(Amount: USD 600)

Support for Pioneering Research Initiated by the Next Generation (JST SPRING) Oct. 2021 – Sept. 2023

Japan Science and Technology Agency (JST)

(Grant No. JPMJSP2136; Amount: JPY 2,400,000 per year)

Graduate School Research Support Scholarship

Kyushu University

(Amount: JPY 250,000)

Sept. 2021

First-Class Scholarship of the Graduate School

Dalian University of Technology

(Amount: RMB 19,600 per year)

Sept. 2018 – June 2020